

1. (Convergence of policy iteration) Given an MDP with a finite state space, action space, and reward set. Prove that a policy improvement step will always produce a new policy at least as good as the original one, and prove that policy iteration converges to an optimal policy.
2. (Convergence of temporal difference value learning) In TD value learning, we repeatedly try out a given policy. At the n th trial, we collect a sampled value x_n and use it to update our value estimate V_n . Suppose we use a time-varying learning rate of $\alpha_n = 1/n^2$ to compute the exponential moving average, and assume that both $\{x_n\}$ and $\{V_n\}$ are bounded sequences, i.e., $\forall n > 0, |x_n| \leq C_1$ and $|V_n| \leq C_2$ for some constants $C_1 > 0$ and $C_2 > 0$. Prove that TD value learning using this particular choice of learning rates $\{\alpha_n\}$ will converge. Hint: Verify that the sequence $\{V_n\}$ with $V_n = (1 - \alpha_n)V_{n-1} + \alpha_n x_n$ is a Cauchy sequence.